THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM ONE
032 CHEMISTRY
MARKING SCHEME AUGUST 2026

SECTION A (15 MARKS)

Question 1

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	A	B	B	B	C	B	A	B	C

(10 marks)

Question 2

(i)	(ii)	(iii)	(iv)	(v)
B	D	A	C	E

(5 marks)

SECTION B (70 MARKS)

Question 3

(a) Chemistry: The branch of science that deals with the study of matter, its properties, composition and the changes it undergoes.**(2 marks)**

(b) **Four ways chemistry contributes to society — (8 marks)**

(i) **Medicine:** Chemistry is used in the manufacture of medicines and drugs for treating diseases.

(ii) **Agriculture:** Chemistry is used in producing fertilizers, pesticides and other agricultural chemicals.

(iii) **Water treatment:** Chemical substances and processes are used to purify water and make it safe for domestic use.

(iv) **Food production/preservation:** Chemistry helps in food processing, preservation and quality control.

(v) **Manufacturing:** Chemistry is applied in production of soaps, detergents, plastics, paints and other products.

(vi) **Environmental protection:** Chemistry helps in controlling pollution and treating waste.

Total: 10 marks

Question 4

(a) Four disciplines related to chemistry — (8 marks)

(i) **Biology:** Chemistry explains chemical substances and reactions occurring in living organisms.

(ii) **Physics:** Chemistry uses physical principles to understand properties and behaviour of matter.

(iii) **Agriculture:** Chemistry helps in understanding soils, fertilizers and pesticides.

(iv) **Medicine:** Chemistry is used to develop and manufacture medicines.

(v) **Pharmacy:** Chemistry is used in preparation, formulation and analysis of medicines.

(vi) **Environmental science:** Chemistry helps in studying and controlling pollutants.

(vii) **Food science:** Chemistry helps in understanding food composition, preservation and processing.

(b) **Two benefits of learning chemistry (2 marks)**

(i) Develops scientific knowledge and problem-solving skills.

(ii) Helps students understand materials and chemical changes in everyday life.

(iii) Promotes safe handling of chemicals.

(iv) Provides knowledge useful in agriculture, medicine and industry.

Total: 10 marks

Question 5

(a) Appropriate apparatus **(5 marks)**

(i) Measuring a known volume of liquid	Measuring cylinder
(ii) Heating a small amount of solid or liquid	Test tube
(iii) Measuring temperature	Thermometer
(iv) Transferring a small amount of solid chemical	Spatula
(v) Observing very small objects	Microscope

(b) Five laboratory rules **(5 marks)**

(i) Do not eat or drink in the laboratory.

(ii) Wear appropriate protective clothing/goggles where necessary.

(iii) Follow the teacher's instructions.

(iv) Do not taste chemicals.

(v) Do not smell chemicals directly.

(vi) Keep the laboratory clean and tidy.

(vii) Report accidents immediately.

(viii) Handle glassware carefully.

(ix) Never perform unauthorized experiments.

Total: 10 marks

Question 6

(a) Three measures **(6 marks)**

(i) Handle glassware carefully.

(ii) Do not use cracked or damaged glassware.

(iii) Clean broken glass using a brush and dustpan/tongs, not bare hands.
(iv) Dispose of broken glass in a designated container.
(v) Keep glassware away from the edge of benches.

(b) Two first-aid actions **(2 marks)**
(i) Cool the affected area under clean running water.
(ii) Remove nearby tight items such as jewellery if it can be done safely.
(iii) Cover the burn with a clean sterile dressing.
(iv) Report the injury to the teacher/first-aider.
(v) Seek appropriate medical attention where necessary.

(c) Two reasons for reporting accidents immediately **(2 marks)**
(i) To receive immediate first aid/treatment.
(ii) To prevent the accident from causing further injury.
(iii) To enable the teacher/laboratory technician to make the area safe.
(iv) To ensure the accident is properly recorded and investigated.

Total: 10 marks

Question 7

(a) Three conditions necessary for fire **(3 marks)**
(i) Heat/ignition source
(ii) Fuel/combustible material
(iii) Oxygen/air

(b) Three suitable fire-extinguishing methods **(3 marks)**
(i) **Water** — removes heat/cools the burning material.
(ii) **Sand** — covers the fire and cuts off oxygen.
(iii) **Carbon dioxide extinguisher** — excludes oxygen from the fire.
(iv) **Fire blanket** — covers the fire and cuts off oxygen.
(v) **Foam extinguisher** — forms a layer that separates the fuel from oxygen.

(c) Two precautions when dealing with laboratory fire **(4 marks)**
(i) Inform the teacher/laboratory technician immediately.
(ii) Evacuate to a safe place.
(iii) Use the appropriate fire extinguisher.
(iv) Do not panic or run unnecessarily.
(v) Do not fight a large/uncontrollable fire.
(vi) Switch off gas/electricity if this can be done safely.

Total: 10 marks

Question 8

(a) Types of flames **(2 marks)**
(i) Yellow flame — **Luminous flame**
(ii) Blue flame — **Non-luminous flame**

(b) Three differences **—(6 marks)**

Yellow/Luminous flame	Blue/Non-luminous flame
(i) Yellow in colour	(i) Blue in colour
(ii) Produces soot	(ii) Produces little or no soot
(iii) Incomplete combustion	(iii) More complete combustion
(iv)Less hot	(iv) Hotter
(v)Less suitable for heating	(v) More suitable for heating
(vi) Usually formed when air supply is restricted	(vi) Formed when sufficient air is supplied

(c) Two situations where a non-luminous flame is preferred (2 marks)

(i) Heating substances in the laboratory.

(ii) Sterilising/strong heating of suitable laboratory equipment.

(iii) Any situation requiring a hot, clean flame with little/no soot.

Total: 10 marks

Question 9

(a) Three states of matter **(3 marks)**

(i)Solid (ii) Liquid (iii) Gas

(b) Three differences **(6 marks)**

(i) **Shape:** Solids have a definite shape; liquids take the shape of their container; gases have neither definite shape nor definite volume.

(ii) **Volume:** Solids and liquids have definite volume; gases have no definite volume.

(iii) **Particles:** Solid particles are closely packed and vibrate in fixed positions; liquid particles are less closely packed and move past one another; gas particles are far apart and move freely/randomly.

(c) Changes of state (1 **mark)**

Ice → water = **Melting**

Total: 10 marks

SECTION C (15 MARKS)

Question 10

(a) Distinguish between physical and chemical change **(8 marks)**

Physical change	Chemical change
No new substance is formed	New substance(s) are formed
Usually reversible	Usually difficult to reverse
Usually involves change in state, size or shape	Usually involves change in chemical composition
Properties of the substance remain essentially unchanged	Properties of the new substance differ from the original
Usually does not involve a permanent chemical change	Involves a chemical reaction

(b) Classification and reasons **(6 marks)**

(i) Ice melting

- Physical change
- No new substance is formed; water can change back to ice.

(ii) Water boiling

- Physical change
- Only the state changes from liquid to gas; no new substance is formed.

(iv) Iron gate rusting

- Chemical change
  - A new substance, rust/iron oxide, is formed.

(c) Any relevant (example, Piece of paper burning)**(1 mark)**

Total: 15 marks